

Manual

Split Operations and Merged Operations BDE-SSG 8.1

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Contents

1	Übersicht Splitt- und Sammelarbeitsgänge.....	4
2	Splitting Operations.....	5
3	Advanced Split Function	10
	Splitting an operation.....	11
	Cancel split operation.....	15
	Displaying split operations.....	16
	Posting to split operations	16
	Uploading to the PPS/ ERP system.....	17
4	Processing of Merged Operations.....	18
4.1	Creation of merged operations at the terminal.....	18
4.1.1	Different types of posting merged operations.....	19
4.1.2	Posting of merged operations at the terminal.....	19
4.1.3	Posting of merged operations created at the terminal.....	20
4.1.4	Further notes on the generation of merged operations at the terminal	23
4.2	Generation of merged operations in MOC	24
4.2.1	Creation of merged operations in MOC	24
4.2.2	Processing of merged operations (created in MOC)	24
4.2.3	Additional information	26
5	Generate and Dissolve Merged Operations.....	28

1 Übersicht Splitt- und Sammelarbeitsgänge

Purpose

This function package provides functions that can be used to split operations or to consolidate operations and also the posting functions based on the results.

Implementation considerations

You use the function package if:

- You would like to combine different short-running operations in order to be able to enter them "one-to-one".

or if

- You want to shorten operation lead times or better utilize workplaces / machines available for production by processing operations in parallel.

Integration

The functions contained in this package are integrated into different function groups in business data collection. Emphasis here is on data entry, i.e. posting at the terminal, and the posting resulting from it.

Features

- Split operations
 - User dialog for generating and managing split operations
 - Staging of split operations for MES planning and entry in order to plan the latter in parallel and to be able to post quantities and times to them.
 - Merging production activities to the original operation and uploading to the ERP
- Merged operations
 - Function in the MOC for combining operations with short processing duration into merged operations
 - User dialog at the Windows terminal for generating, posting and managing merged operations
 - Staging of merged operations for MES planning and entry
 - Posting times and quantities proportionally to the original operations based on configuration and confirmation

2 Splitting Operations

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Purpose

You use this function if you want to shorten the lead time of operations or improve utilization of several workplaces/ machines available for production.

Integration

This function splits an operation into two or more operations. An operation can be split using the following functions:

- [Order sequencing](#).
- [Edit operations](#).
- [Graphic planning](#): can be called up via the context menu of the operation.

Data is always uploaded to the PPS/ ERP system according to the order/ operation number of the original operation.

Requirements

An operation can only be split if the following conditions are met:

- The length of the split number is set to 1 (maximum: 9 splits) or 2 (maximum: 99 splits) in the HYDRA basic parameter settings.
- The identifier "May be split" is set at the operation.
- The operation to be split cannot be a merged operation.
- The operation is not a split operation. It is not allowed to split a split operation once more.
- The operation has not yet started, i.e. it must be marked with status (control indicator) "prepared".

Splitting an operation

Function authorization	op.split
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The function "Split operation"  allows you to split the total production quantity of an operation into several operations. The number of operations subjected to splitting is predefined as per the number of splits configured in the order backlog. If the indicator "May be split" is not set for the operation, you cannot split the operation.

The system runs through the following processing steps when splitting an operation:

Checks

- A check is run to verify whether the operation may be split. If the indicator "May be split" is set for the operation, you can split the operation.
- The number of splits entered must be greater than 1.
- The entered number of splits must be less than the maximum number of splits defined at the operation.
- The operation to be split may not have already been split (split master).
- The operation to be split may itself not be an operation split.
- The operation to be split may not be a merged operation.

If these conditions are met, the operation may be split.

Processing

The system issues the numbers for the splits automatically by distributing a consecutive number. The number is shown in the field "split": The remainder of the number (order/ operation) matches the one of the original operation ("split master"). The number of splits depends on the length/number of characters defined for splits in the HYDRA basic parameter settings.

Example:

Operation 47100100 0100 is divided up into to three splits; the number length for the split defined in the basic settings is 2.

Results of splitting:

Split master: 47100100 0100 (can no longer be logged on)

Individual splits: 47100100 0100 01

47100100 0100 02

47100100 0100 03

When an operation is split, the following values of the original operation are distributed to the splits based on the number of splits:

- Target quantity (primary, secondary, tertiary, base quantity unit)
- Target scrap (primary, secondary, tertiary, base quantity unit)
- Processing time

The quantities are generally divided up in full numbers. If they cannot be distributed evenly, the remainder is allocated to the last operation split.

All other data is transferred from the original operation.

After splitting, the order is flagged for lead time scheduling.



After splitting, only the generated operation splits can be posted. The original operation (split master) is marked as *locked* and cannot be posted. It is no longer shown in the backlog of orders.

The HYDRA Shop Floor Scheduling module no longer shows the split master and, therefore, the master cannot be planned any longer, but only its split operations.

Changes to operation splits, in particular changes in planning (e.g. changes to planned dates or changes to the planned workplace) do not affect the split master.

Please keep in mind that after an operation has been split, any new changes from the ERP system to this original operation ("split master") will have no effect on the individual operations split off.

If the target quantity update is enabled, the operation splits will not be updated, but only the split master.

Cancel operation splits

Function authorization	op.splitrelease
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When an operation split is canceled, all of the operations split off are deleted and only the original operation, which up until now was the split master, continues to exist. The split master is now treated as a normal operation.

Operation splits can only be canceled if all splits of the operation still show a status "prepared".

Displaying operation splits

Whether split operations are displayed depends on the application:

- The original operation (split master) is not shown in graphic planning.
- The split master is shown in the following applications:
 - Edit operations
 - Order information

Whether the single operation splits are displayed also depends on the application:

- The [graphic planning](#) generally shows operation splits. In contrast to this, the split master is no longer shown after splitting.
- The single operations split off are shown in the following applications, provided that the relevant option has been selected:
 - [Edit operations](#)
 - [Order information](#)

Posting to operation splits

Where split operations are concerned, the log record is ONLY generated for the operation split; no log record is generated for the split master itself.

Only the split master's status is updated if the following events occur:

- interrupt operation
- log off operation

Uploading to the PPS/ ERP system

The values recorded or posted to operation splits are generally uploaded to the PPS/ ERP system according to the split master.

Splitting of operations subject to batch management

General

The "split operation" function may also be run for operations subject to management in batches. The function is identical to the split function of conventional operations.

Procedure

The user applies the "split OP" function and splits off an OP from the "split master" in the given operation quantity (primary quantity).

The operation is split. Data relating to material is transferred from the split master to the operations split off.

The required quantity is calculated subject to the input quantity and the quantity split off.

Results after splitting the operation:

With regard to the material, the following data/ values are accepted or used by the operations split off:

- Order header data
 - Finished article (number of the finished article) --> taken over from split master
 - Article name --> taken over from split master
 - Batch number --> taken over from split master (for the batch being generated in this order, e.g. transferred from the ERP; identical for all operation splits)
 - Material type of the finished article --> taken over from split master
- OP data
 - Material number of output material/article --> taken over from split master
 - Material type of output material/article --> taken over from split master
 - Name of output material/article --> taken over from split master
 - Indicator "batch management requirement" --> taken over from split master
 - Indicator "serial numbers required" --> taken over from split master
- Component list
 - All data included in the component list is copied or transferred from the split master.
 - The required quantity is recalculated from the OP split off based on the quantity split off.
- Output batch change:
 - When output batches are changed, the number of the split master is added to the output batch.
- Uploads to the ERP system:
 - Uploads to the ERP system are always performed in relation to the split master.

3 Advanced Split Function

Usage

You use the split function if you would like to shorten the lead time of operations or you would like to improve the capacity utilization of several workplaces/ machines available for production.

In contrast to the simple split function, the enhanced split function provides the following features:

- An operation may be split into split operations having different target quantities and setup times.
- An operation that has already been split (split operation) may be split into further split operations with different target quantities and setup times.
- Individual splits may be deleted.

Integration

This function provides the ability to split an operation into two or more split operations. An operation can be split in the following functions:

- [Order sequencing](#).
- [Edit operations](#).
- [Graphic planning](#) (can be called up via the context menu at the operation).

Data is always uploaded to the PPS/ ERP system in relation to the order/ operation number of the original operation.

Prerequisite

An operation can only be split if the following conditions are met:

- The length of the split number is set to 1 (maximum 9 splits) or 2 (maximum 99 splits) in the [basic parameter settings](#).
- The option "enhanced split function" has to be set to "yes" in the [basic parameter settings](#).
- The flag "may be split" is set for the operation and the field "max. number of splits" includes a value greater than 2.
- The operation to be split is not a merged operation.



Machines with MPL/TRT functions do not support split operations.

Splitting an operation

Function authorization	op.split
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An operation may be split by using the function "split operation".

To perform splits, the relevant operation is to be selected and the split function is to be started using the context menu or by clicking the relevant button. A dialog opens where the planner may define into how many operations the operation is to be split and what quantities are planned for each split operation.

Field description of "split operation"

The dialog shows different information depending on whether the operation is split for the first time or if splitting is performed for an operation that has already been split.

A note is added to the below fields indicating whether they are available if the OP is

- (1) split for the first time or
- (2) if it has already been split

MES order number (1) (2)

Operation that is to be split. This is the operation that may be split (the split master) if splitting is performed for the first time. This is the split master for further splitting.

Target quantity (P) (1) (2)

Target quantity and unit of the split master according to the specifications made in the pool of orders. The target quantity and the below-mentioned quantities generally refer to the primary quantity of the operation.

Actual quantity (1) (2)

(Yield) quantity that has been uploaded/confirmed so far. This quantity is entered by the split operations and carried forward at the split master status.

It includes the actual quantity of operations that have already been finished as well as the actual quantity of running or interrupted operations.

Remaining quantity for split OPs (1) (2)

Refers to all split OPs that have not yet been finished (split OPs with control indicator S, V, L or U) with the difference resulting from "target quantity minus actual quantity" being > 0: total of remaining quantities

Operations with a difference between target quantity and actual quantity that is less than 0 are not taken into account, as this resulted in an overproduction and this surplus is no longer produced and, as a result, no longer considered for further splitting.

Difference (1) (2)

This quantity results from:

"Target quantity" (P) minus "actual quantity" minus "remaining quantity of split OPs"

If this quantity is unequal to 0, the note "DIFFERENCE" is shown in red. This shows whether/when the entire target quantity of the split master is completely distributed.

Max. number of splits (1) (2)

The maximum number of splits defined for the split master

Previous number of splits (1) (2)

Number of splits that have been performed so far. This number is the total of the split operations and the splits to be performed now (splits included in the list)

Remaining run time (1) (2)

Remaining run time of the split master according to the remaining run time formula.

Setup time (1) (2)

Target setup time of the split master according to the specifications made within the pool of orders.

The setup time is taken over into the list during splitting and can be adjusted there individually for each split.

The information listed above generally refers to the operation to be split or to the split master, provided that the operation has already been split.

In case the operation has already been split, i.e. it is now the split master, the below-mentioned information is shown for the split operation that is to be used for further splitting. If the operation is split for the first time, these fields will *not* be displayed:

Split OP (2)

Split operation that is to be split or that is to be used for further splitting.

Target quantity (P) (2)

Target quantity of the split operation

Actual quantity (2)

(Yield) quantity uploaded/confirmed so far for this split operation

The planner's interactive input options

The splits are generated in the list, once the "number of splits" and, if necessary, the "quantity to be split" have been entered and the "split" button has been clicked. The target quantity as well as the setup time can still be changed.

Number of splits (1) (2)

Planners can now specify how many splits they want to create

By default, the field is assigned the number of splits that are still possible (maximum number of splits minus current number of splits).

Quantity to be split (1) (2)

In addition, the planner can define a split quantity. In this case, the individual splits are assigned this quantity in the table

If the quantity to be split is 0, the remaining quantity will be spread evenly among the splits. However, the quantity may still be changed for each split.

But splits are only added until the maximum number of splits of the split master has been reached.

Generate OP with difference quantity (1)

This checkbox is only visible, provided that the OP has not yet been split and is not set by default.

If the checkbox is enabled, another split operation is shown in the table in addition to the split OPs, provided that the total target quantity of the split OPs to be generated (included in the list) is less than the remaining quantity of the operation to be split.

All target data, except for the target quantity itself, may be changed for this operation. The target quantity cannot be changed, as it is calculated automatically: remaining quantity between the operation and the total target quantity of the split OPs to be generated.

Set off target qty. of the split OP (2)

This checkbox is only visible, provided that it is a split operation and it is not set by default.

If this checkbox is enabled, the target quantity of the split OP is updated, i.e. the target quantity of the split OP is reduced by the total target quantity of the split operations to be generated. However, this is only the case if this sum does not exceed the remaining quantity of the split operation (the remaining quantity results from the difference between target quantity and actual quantity). Otherwise, the checkbox cannot be selected.

Table columns

Split (1) (2)

The system generates the future split number. It cannot be changed.

The next free number is used as the split number, i.e. the current maximum split number is verified.

Target quantity (P) (1) (2)

Target quantity of the split operation. The planner may change the target quantity, provided it is not the split generated due to the option "Generate OP with difference quantity".

The target quantity of new entries results from the "quantity to be split".

Setup time (1) (2)

Target setup time of the split operation. The planner may change the setup time.

New entries are assigned the setup time of the split master.

Remaining run time (1) (2)

Remaining run time (remaining processing time) according to the remaining run time formula that is defined for the split master taking into account the target quantity of the split operation. HYDRA calculates this remaining run time that cannot be changed by the planner.

The totals row shows:

- Column "Split": number of entries,
- Column "Target quantity": Total of the target quantity,
- Column "Setup time": Total of setup times,
- Column "Remaining run time": Total of remaining run times.

An error message is output if the entered number of splits (existing and new splits) exceeds the maximum number of splits.

It is allowed to enter an (admissible) number of splits and a split quantity that is greater than the target quantity of the operation. However, a respective note saying "difference" is displayed for the user.

Editing of split entries

Existing entries can be edited or deleted using the "edit" or "delete" button. Multiple selections are supported for deletion.

Once entries have been confirmed by clicking OK, they are updated in the relevant dialog.

The target quantity cannot be changed if the checkbox "Generate OP with difference quantity" is set.

Split result

The entries are checked, once the dialog "split operation" has been confirmed by clicking OK. Error messages are shown if necessary:

- Splits having a target quantity 0 are not allowed.
- In general, the maximum number of splits is checked against the split master.

Then the system generates the split operation(s). They are directly generated in the database.

One or several new operations are generated by splitting. Their ID differs from the original split master by the split number. There are the following operations, for example:

- 123456789012 0100 = Split master
- 123456789012 0100 1 = Split operation 1
- 123456789012 0100 2 = Split operation 2

Please also note



After splitting, only the split operations that were created can be posted. The original operation (split master) is identified as *locked* and can therefore not be posted. It is neither shown in the pool of orders any longer.

The split master will neither be shown any longer in the HYDRA Shop Floor Scheduling module and, as a result, it can no longer be planned, but only its split operations.

Changes to split operations, in particular changes in planning (e.g. changes to planned dates or changes to the planned workplace) do not affect the split master.

Please keep in mind that after an operation has been split, any new changes from the ERP system to this original operation ("split master") will have no effect on the individual split operations.

If the target quantity update is enabled, the split operations will not be updated, but only the split master.

Cancel split operation

Function authorization	op.splitrelease
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A split operation can be canceled by clicking the button , provided that it has not yet been started.

In contrast to the simple split function, only the selected split operation is canceled, i.e. deleted for the advanced split function. The split master and the other split operations remain unchanged.



Please note: If a split operation is deleted, its target quantity will not be transferred to other, still existing split operations.

If the last split operation is deleted, the split master becomes a "normal" operation.

Displaying split operations

Whether or not split operations are displayed depends on the context of the relevant application.

Split master:

- The original operation (split master) is not shown in graphic planning.
- The split master is shown in the following applications, among others:
 - Edit operations
 - Order information

Split operations:

- The [graphic planning](#) generally shows split operations. In contrast to this, the split master is no longer shown after splitting.
- Split operations are shown in the following applications, provided that the relevant option has been selected:
 - [Edit operations](#)
 - [Order information](#)

Posting to split operations

Where split operations are concerned, the log record is ONLY generated for the split operation; no log record is generated for the split master itself.

Only the split master's status is updated if the following events occur:

- interrupt operation
- log operation on

Uploading to the PPS/ ERP system

The values recorded or posted to the split operations are generally uploaded to the PPS/ ERP system in relation to the split master.

4 Processing of Merged Operations

Utilization

Merged operations are a special form of serial production. Different short-duration operations are combined into logical blocks with manageable runtimes (so-called "merged operations") within the planning level (e.g. in HYDRA shop floor scheduling) or at shop floor terminals. The system creates a "representative" operation for these merged operations, which is logged on for all the individual operations contained in the group. The entered data is divided according to different configurable perspectives.

Integration

HYDRA supports two methods of creating merged operations:

Creation of merged operations at the terminal

The merged operation is created by the worker at the terminal. After logging the merged operation on, all the concerned individual operations are specified one after the other and assigned to the concerned person. Only the person needs to be specified while logging out or interrupting the merged operation.

Creation of merged operations in MOC

The function for creating merged operations combines the individual operations into a representative OP. The representative operation is then logged on to the terminal.

In contrast to merged operations at the terminal, merged operations that are created in MOC can be planned using the scheduling functions provided in MOC.

Prerequisite

Creating merged operations at the terminal is incompatible with creating merged operations in MOC. Hence, the method to be used must be defined in the *HYDRA basic settings* through appropriate configuration. Subject to this configuration, the functions for merged operations are available either at the terminal or in MOC only.

4.1 Creation of merged operations at the terminal

The "merged OP at terminal" function makes it possible to combine several operations into a single merged operation (MOP).

This is helpful if:

- several short-duration operations can be combined into a single operation (less posting efforts),

- several operations are “at the workplace” simultaneously, so that the logon/logoff postings concerning the individual operations cannot otherwise be sent in the correct order (for example during annealing or tempering in the metal industry or smoking in the F&B sector).

4.1.1 Different types of posting merged operations

The different types of posting merged operations, which are described below, are configured in the [terminal configuration](#).

"Generation per person"

MOP logons are combined person-based order logons, i.e. the person who combines several individual operations into an MOP is logged on at all individual operations automatically.

Only one MOP is possible per person. If a person executes another MOP logon at a later time, then these individual operations are automatically assigned to the already existing MOP logon.

An MOP can only be logged on, logged off or interrupted. The logging on of additional persons to a personal MOP is impossible.

"Generation per machine"

Several OPs can be grouped in a single merged operation per machine/workplace on the terminal using the “log merged OP on” function. With one MOP permitted per workplace.

Just as it is the case for individual operations, several people may log on to this MOP. If the MOP is interrupted or logged off all registered people will be logged off automatically. Posting of quantities and splitting of times is made in accordance with the specifications described in the sections that follow.

4.1.2 Posting of merged operations at the terminal

The method of creating merged operations at the terminal can be taken from the concerned terminal documentation (*for Windows terminals or DOS terminals*).

Merged operations are created at the terminal as described in the corresponding documentation. The terminal generates a representative order number for the merged operation. Subject to the posting type, the user's badge number or the machine/workplace number are integrated in this order number:

Merged operation number for the "generation per person" option

Depending on the length of the HYDRA order no. and the length of the staff badge no. configured in the system, the number of the merged operation is created as follows:

The length of the order no. is at least four characters longer than the staff badge no.:

SAM-XXXXX xxxxx represents the staff badge no.

The length of the order no. is at least two characters longer than the staff badge no.:

S-XXXXX xxxxx represents the staff badge no.

The order number length is **not** at least two characters longer than the staff badge number:



In this case, the standard function "merged operations at the terminal" cannot be used.

Merged operation number for the "generation per machine" option

Depending on the length of the HYDRA order no. and the length of the machine number configured in the system, the number of the merged operation is created as follows:

Length of the order number is at least two characters longer than the length of the machine/workplace number:

S-XXXXX xxxxx represents the machine/workplace number

Length of the order number is at least four characters longer than the length of the machine/workplace number:

SAM-XXXXX xxxxx represents the machine/workplace number

The length of the order number is **not** at least two characters longer than the length of the machine/workplace number:



In this case, the standard function "merged operations at the terminal" cannot be used.

4.1.3 Posting of merged operations created at the terminal

The entered actual times of the MOP can be divided according to the following methods

According to the **standard times**, i.e. in relation to the standard times of the individual OPs

According to the **default quantities**, i.e. in relation to the target quantities of the individual OPs

According to the **individual OPs**, i.e. according to the number of the logged on individual OPs

The posting method is defined in the [basic parameter settings](#).



The options "according to standard time" or "according to default quantity" must not be used together with the option "Proportionate RPA posting in personnel postings" in the basic parameter settings.



Please contact MPDV Support to configure the MOP posting method.

Example - Distribution according to standard time

Example of dividing the real times of the MOP in relation to the individual standard times of the individual OPs:

3 operations are combined into one MOP:

OP	Standard time
OP01	8000 Sec.
OP02	600 Sec.
OP03	6000 Sec.

A log record for the MOP with the runtime of 12000 sec. must now be posted onto the individual OPs. The times are to be divided according to the following formula

$$\text{Posted times} = \frac{\text{Total duration of the log record} * \text{standard time of individual OP}}{\text{Total of the standard times of all OPs}}$$

The following values of the individual OPs can be derived from the formula:

OP	Posted runtime
OP01	6575 Sec.
OP02	493 Sec.
OP03	4932 Sec.

Calculation example for OP01:

Sum of standard times of all individual OPs = 8000 + 600 + 6000 = 14600

Posted runtime = 12000 * 8000/14600 = 6575 Sec.

PLEASE NOTE:

The standard time of an operation is calculated from the **target setup time + target duration**.

Example - Distribution according to standard time

Example of dividing the actual times of the MOP in relation to the individual target quantities of the individual OPs:

3 operations are combined into one MOP:

OP	Target quantity
OP01	1000
OP02	500
OP03	2500

A log record for the merged OP with the runtime of 12000 sec. must now be posted onto the individual OPs. The times are to be divided according to the following formula

$$\text{Posted time} = \frac{\text{Total duration of log record} * \text{target quantity of individual OP}}{\text{Total of target quantities of all individual OPs}}$$

The following values of the individual OPs can be derived from the formula:

OP	Posted runtime
OP01	3000 Sec.
OP02	1500 Sec.
OP03	7500 Sec.

Calculation example for OP01:

Sum of target quantities of all individual OPs = 1000 + 500 + 2500 = 4000

Posted runtime = 12000 * 1000/4000 = 3000 Sec.

Example – Distribution according to the number of individual operations

Example of dividing actual times of the MOP according to individual OPs

3 OPs are combined in one MOP:

A log record for the MOP with the runtime of 12000 sec. must now be posted onto the individual OPs. The times are to be divided according to the following formula

$$\text{Posted time} = \frac{\text{Total duration of the log record}}{\text{number of individual OPs}}$$

The following values of the individual OPs can be derived from the formula:

OP	Posted runtime
OP01	4000 Sec.
OP02	4000 Sec.
OP03	4000 Sec.

The **quantities as well as the entered times** are also **posted for the person** who edits the MOP.

4.1.4 Further notes on the generation of merged operations at the terminal

The following particularities apply while processing merged operations at the terminal

- By default, the "merged operations" function is not active and must be enabled with respect to a specific terminal in the terminal configuration .
- Only the ADE terminal provides MOP functions at DOS terminals . However, Windows terminals also provide MOP functions if the "MDE" operation mode is used.
- To complete an MOP, the "logon of several OPs" option must be set in the workplace / machine configuration, i.e. the logging on of several operations at this workplace is permitted.
- The "max. OPs per person" option of the HR master is also taken into account when it comes to the creation of merged operations on the terminal. Consequently, a person may only log one operation on as merged operation if the "max. OPs per person" option is set to 1.
- A person can combine a maximum of 20 individual operations into a merged operation.
- In MOP log on, no sequencing list is supported at DOS terminals , i.e., the order numbers are to be entered with the help of barcodes or through manual entry.
- For merged operations that are logged on without recording of number of pieces, the produced actual quantity of the individual OPs is set to the respective target quantity of pieces at the time of logoff.
- With merged operations that are logged on with the entry of quantities, each operation receives its specified quantity, when the merged operation is interrupted or logged off.
- For merged operations that are logged on to workplaces/machines with automatic quantity entry, each individual operation gets its own proportional quantity. The quantities are divided according to the same key as it is the case for times. This division is described in the sections that follow.
- It is sufficient to enter the badge number, when merged operations are logged off/interrupted (MOP person). Thus, all operations of a person's merged operation are logged off automatically.

Waiting period processing

In case an MOP logon has to be dated forward due to waiting period processing, the first individual operation of this merged operation is dated forward. Further OPs that might be added to the MOP at a later time are not affected.

An MOP or all individual operations pertaining to it are automatically interrupted when the person who processes it/them clocks out.

Provided that the waiting period processing function is configured accordingly, it is also possible that the person who processes the MOP is automatically logged in again as soon as they clock-in in HYDRA-PZE.

4.2 Generation of merged operations in MOC

This function enables the combination of individual operations into merged operations in MOC.

The functions for creating or canceling merged operations are available in the "order overview" as well as in the "order sequencing" dialog (the BDE-FST license is a precondition here).

If a merged operation is created in MOC, then the "members" of a merged operation cannot be viewed here anymore. The same is true of the sequencing list at the terminals.

The number of members in a merged operation is not limited. However, only prepared orders/OPs that are not already contained in other merged operations can be added to merged operations.

Merged operations themselves cannot flow into other merged operations.

The logging on of a merged operation at the terminal takes place as representative for all the members of a merged operation. Merged operations can be logged on, interrupted or logged off at the terminal the same way as normal operations.

4.2.1 Creation of merged operations in MOC

The method of creating merged operations is described [here](#).



4.2.2 Processing of merged operations (created in MOC)

While dividing the entered quantities and times of a merged operation among the associated individual operations, the system supports the "homogeneous" (not for merged operations created at the terminal) and the "non-homogeneous" method.

Homogeneous merged operations

In a homogeneous merged operation, quantities and times are divided on the basis of the overrun principle, i.e. all the individual operations are "filled up" one after the other according to the specified target number of pieces. If the production surpasses the quantity planned for the total merged operation, then this excess and the corresponding times are added to the individual operations having the largest order number.

Simplified example:

OP	Target quantity	Target run time
OP01	200	4.0 hours
OP02	450	9.0 hours
OP03	250	4.5 hours

A log record of the merged operation containing the real quantity 500 and real duration 5.0 hours must now be “re-distributed” over the individual operations. The number of pieces is posted to the individual OPs one after the other (up to the specified target quantity). Depending on the posted number of pieces, the time specifications are then calculated on the basis of the following formula:

$$\text{Posted time} = \frac{\text{Total log record time} * \text{posted quantity}}{\text{Total quantity of the log record}}$$

The following values of the individual OPs can be derived from the formula:

OP	Posted quantity	Posted runtime
Individual OP01	200	2.0 hours
Individual OP02	300	3.0 hours.
Individual OP03	0	0.0 hours

Non-homogeneous merged operations

The quantities and times are posted proportionally among all the individual operations in a **non-homogeneous merged operation**. If excess production takes place, then the excess is divided among all the individual operations.

Simplified example:

OP	Target quantity	Standard time
Individual OP01	400	8.0 hours
Individual OP02	20	0.5 hours
Individual OP03	30	1.0 hours

A log record for the merged operation with data real quantity 200 and runtime 4 hours must now be “re-distributed” among the individual operations. The following formula should be used in this context:

Division of times

$$\text{Posted time} = \frac{\text{Standard time of individual OP} * \text{total of standard times of all individual OPs}}{\text{Total duration of the log record}}$$

Division of quantities

$$\text{Posted quantity} = \frac{\text{Target quantity of individual OP} * \text{total target quantities of all individual OPs}}{\text{actual quantity of the log record}}$$

The following values for the individual OPs can be derived from the formula:

OP	Posted quantity	Posted runtime
OP01	178	3.4 hours
OP02	9	0.2 hours
OP03	13	0.4 hours

Calculation example (for individual OP01)

Posted runtime = $8/9.5 * 4 = 3.4$

Posted quantity = $400/450 * 200 = 178$ pieces.

PLEASE NOTE

The standard time of an operation is calculated from the **target setup time + processing time**. All data for this calculation is located in the order backlog of the operation.

The definition of the merged operation type is undertaken homogeneously on a one-time basis through *HYDRA basic settings* for the entire HYDRA system with the help of the "processing of merged operations" option.

The upload of the **entered times** to the PPS system takes place according to the definition of the merged operation type – homogeneous or non-homogeneous.

If a member of a merged operation e.g., is closed through the ERP interface, then it is left out of the above mentioned settlements.

4.2.3 Additional information

Validation checks

As a rule, validation checks performed as part of the posting process (e.g. target quantity checking for partial uploads) are only made for the merged operation, not for the individual operations assigned to the merged operation.

Modifications in OP data

Modifications of MOPs

Modifications in MOP do NOT lead to changes in the individual OPs.

Changes to a member of a merged operation

Modifications to a member of a merged operation do not affect the master operation.

Here too, whether or not modifications are permitted generally depends on its status, as in the case of a normal operation.

5 Generate and Dissolve Merged Operations

Summary

Menu	Production control → Production overview → Operations Production control → Production support → Pool of orders Production control → Production support → Order sequencing	
Function authorization	op.colopcreate op.coloprelease	Generate merged operation Dissolve merged operation

Usage

The function used to generate merged operations consolidates the individual OPs to one representative OP. The representative OP is then logged on at the terminal.

Integration

As opposed to merged operations generated at the terminal, merged operations that were generated at the console can be planned using the planning functions in MOC.

Information relating to processing and posting merged operations can be found [here](#).

Requirement

The ability to generate merged operations in MOC requires that the relevant configuration is active in the *HYDRA basic settings*. Depending on this configuration, the merged operation functions can either only be used at the terminal or in MOC.

Generating a merged operation

To generate a merged operation, you must select all of the operations that you would like to include in the merged operation. To select several operations (or also to deselect them), hold down the Ctrl key while clicking on the separate operations.

The screenshot shows a software window titled "Generate merged operation". It features a "Main page" tab and a form with the following fields:

- MOP order number: [] ...
- OP: 0000
- Target quantity (P): 1,520
- Target scrap quantity (P): 0
- Setup time: 1:00:00 Hrs
- Processing time: 20:00:00 Hrs
- Reference OP: 530000810100 (dropdown menu)



By pressing the button , the function "Generate merged operations" is called up. A dialog opens that has the following input fields:

MOP order number: Enter the order number here under which you would like to create the merged operation. If you would prefer that HYDRA issues the number automatically, leave this field empty.

Operation: By default, the system suggests operation number 0000 here.

Target quantity: By default, the sum total of all target quantities (primary quantity unit) of each of the separate operations is suggested as the target quantity. Please keep in mind that no unit conversion is performed here and as such the operations should all have the same primary quantity unit.

Setup duration: By default, the maximum among the highlighted operations is suggested as the setup duration.

Processing time: By default, the sum total of all processing times of each of the separate operations is suggested as the processing time.

Reference OP: The data from the order/ operation defined in the field Reference OP ("master copy") are pulled for the merged operation. The selected operation with the lowest order/ operation number is used for the pre-assignment.

After confirming the dialog, the new merged operation is generated.



Changes to the merged operation DO NOT affect the stock data that are to be incorporated into this merged operation.

When the merged operation is dissolved into its "members", these still contain the data that they had at the time the merged operation was generated.

Dissolving a merged operation

As long as it has not yet been started (status "V"), you can still break down a merged operation into its



"Members" again by highlighting it and by then pressing the  button. You will no longer be able to dissolve a merged operation after it has started.



Changes to the merged operation DO NOT affect the stock data that are to be incorporated into this merged operation.

When the merged operation is dissolved into its "members", these still contain the data that they had at the time the merged operation was generated.

Displaying merged operations

You can use the selection feature in the evaluations/ overviews to control in how far the members of merged operations or a single merged operation are displayed.

Indiv. OPs

Only "normal" operations are displayed, no merged operations and also no members of merged operations

Merged operations

Only merged operations are displayed (merged operation masters)

Indiv. OPs summarized in merged OP

Only MOP members are displayed